Question	Answer	Marks
1	**Defining the problem**	
	vary *m* and measure *v* **or** *m* is the independent variable and *v* is the dependent variable	**1**
	keep *h* constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • axle resting on support(s) (on stands) • supports placed on bench • light gate (connected to timer) positioned at a distance *h* • light gate labelled and *h* indicated	**1**
	vertical metre rule clamped to a stand in a position close to block to measure *h*	**1**
	method to determine *v* using an interrupt length, e.g. *v* = length of block / time recorded by the timer	**1**
	method to measure *m*, e.g. use a (top-pan) balance	**1**
	Method of analysis	
	plot a graph of $\frac{1}{v^2}$ against $\frac{1}{m}$ or equivalent Do not accept logarithms.	**1**

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Question	Answer	Marks
1	<table><tr><td>$\frac{1}{v^2}$ against $\frac{1}{m}$</td><td>$\frac{1}{m}$ against $\frac{1}{v^2}$</td></tr><tr><td>$P = \frac{1}{h \times y\text{-intercept}}$</td><td>$P = -\frac{\text{gradient}}{y\text{-intercept} \times h}$ **or** $P = \frac{\pi r^2 z \times \text{gradient}}{2Qh}$</td></tr></table>	**1**
	<table><tr><td>$\frac{1}{v^2}$ against $\frac{1}{m}$</td><td>$\frac{1}{m}$ against $\frac{1}{v^2}$</td></tr><tr><td>$Q = \frac{\pi r^2 z}{2Ph \times \text{gradient}}$ **or** $Q = \frac{\pi r^2 z \times y\text{-intercept}}{2 \times \text{gradient}}$</td><td>$Q = -\frac{\pi r^2 z \times y\text{-intercept}}{2}$</td></tr></table>	**1**
	Additional detail including safety considerations	**6**
	D1 precaution linked to falling block resulting in damage to block / bench, e.g. use a cushion / sand tray to prevent damage to bench **or** precaution linked to stands falling, e.g. clamp stand(s) to the bench prevent stand falling	
	D2 keep *r* **and** *z* <u>constant</u>	
	D3 method to determine *r* e.g. use calipers / ruler to measure diameter *d* **and** $r = d / 2$	
	D4 measure *z* with a micrometer / calipers	

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Question	Answer		Marks
1	D5	set square correctly positioned between rule and bench to ensure that rule to measure *h* is vertical	
	D6	method to keep *h* constant by identifying constant initial position of the bottom of the block, e.g. clamped pin / rod to indicate the starting point each time or (fiducial) marker on rule	
	D7	description of method to ensure that axle can rotate, e.g. axle is lubricated at the supports to enable axle to rotate, axle is not fixed at the supports so axle can rotate	
	D8	use a large length of block to reduce (percentage) uncertainty in interrupt time or increase the time light gate is interrupted	
	D9	repeat experiment for the same value of *m* **and** determine the average *v*	
	D10	relationship valid <u>if</u> a straight line is produced (with *y*-intercept = $\frac{1}{hP}$). Do not accept line passing through the origin.	

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